The Gaussian Schrödinger–Bass Bridge: Explicit Optimizers and the Solvability Boundary

Abstract

We provide a self-contained derivation, via Pontryagin's maximum principle, of the explicit Gaussian solution of the Schrödinger–Bass Bridge (SBB) problem introduced by Henry-Labordère, Loeper, Mazhar, Pham, and Touzi [4] and studied independently in [7, 6]. The SBB interpolates between the classical Schrödinger bridge ($\beta \to \infty$) and the Bass martingale transport ($\beta \to 0$) via a parameter $\beta > 0$. Our Pontryagin approach yields the adjoint equation $\dot{p} = 2p^2$ in closed form, reducing the boundary value problem to a quadratic equation. We identify the *solvability boundary*: the critical target variance $\gamma_T^* = (\gamma_0 + \beta T^2)/(1 + 2\beta T)^2$ below which the Gaussian SBB admits no finite-cost solution. Going beyond the Gaussian setting, we prove that γ_T^* is a *second-moment lower bound* for solvability of the SBB with arbitrary (non-Gaussian) centered marginals. We characterize the degenerate boundary regime, where the optimal diffusion vanishes terminally and the value function has a jump discontinuity. A numerical example with non-commuting covariance matrices demonstrates anisotropic costate coupling in dimensions $d \geq 2$.

1 Introduction

The Schrödinger bridge problem, originating in Schrödinger's 1931 thought experiment [3], seeks the most likely evolution of a cloud of independent Brownian particles conditioned to match prescribed initial and terminal distributions. In modern formulation, given marginals $\mu_0, \mu_T \in \mathcal{P}_2(\mathbb{R}^d)$, one minimizes the relative entropy of a semimartingale law with respect to Wiener measure, subject to the marginal constraints. The Bass construction [2] provides a different paradigm: one seeks a martingale interpolation between two marginals by controlling only the diffusion coefficient of a local martingale.

Henry-Labordère, Loeper, Mazhar, Pham, and Touzi [4] introduced the *Schrödinger–Bass Bridge* (SBB) problem, which interpolates between these two frameworks via a single parameter $\beta > 0$. The SBB simultaneously controls both the drift α_t and the diffusion coefficient $a_t = \sigma_t \sigma_t^\top$ of a semimartingale, penalizing drift via $\frac{1}{2}|\alpha_t|^2$ and diffusion deviations from the identity via $\frac{\beta}{2}\|a_t - I_d\|_F^2$. As $\beta \to \infty$, the diffusion coefficient is forced to I_d and one recovers the classical Schrödinger bridge; as $\beta \to 0$, the drift penalty dominates and one approaches the Bass martingale transport.

Independently, Backhoff-Veraguas, Beiglböck, Bifronte, and Ley [1] studied the martingale Schrödinger bridge from a related perspective, while Hasenbichler, Pammer, and Thonhauser [5] developed a weak transport approach. A PDE derivation of the one-dimensional

SBB system, including explicit solutions via Legendre transforms and the heat equation, was given in [7]; computational aspects were addressed by Alouadi et al. [6]. The Gaussian Schrödinger bridge (without Bass interpolation) has a well-known closed form [8, 3]; generalizations of the Bass martingale to arbitrary reference measures were developed by Tschiderer [9].

A key finding of [4] is that the SBB is well-posed under $\beta T > 1$. The Gaussian case was solved explicitly in [7] via PDE methods. However, the following questions remained open: the precise solvability boundary for Gaussian marginals (and its relationship to the $\beta T > 1$ condition), whether this boundary extends to non-Gaussian marginals, the structure of the degenerate regime at the boundary, and the behavior of non-commuting covariance matrices in $d \geq 2$ dimensions.

Main contributions. We provide a self-contained derivation of the Gaussian SBB via Pontryagin's maximum principle (complementing the PDE approach of [7]) and address the open questions above. Our results are organized as follows:

(i) **Theorem 3.2** gives the explicit optimizer for the one-dimensional Gaussian SBB. The optimal drift and volatility are expressed in closed form through a scalar parameter p_0 determined by a quadratic equation.

(ii) **Theorem 4.1** identifies the *solvability boundary*: the critical target variance $\gamma_T^* = (\gamma_0 + \beta T^2)/(1 + 2\beta T)^2$, below which no Gaussian SBB solution exists. The Gaussian SBB is solvable if and only if $\gamma_T \geq \gamma_T^*$. This threshold clarifies the role of the condition $\beta T > 1$ from [4]: it ensures universal solvability across all marginals, but specific marginal pairs may be solvable even when $\beta T \leq 1$.

(iii) **Theorem 5.1** extends the results to d dimensions, where the adjoint equation becomes the matrix Riccati equation $\dot{P} = 2P^2$ with solution $P(t) = P_0(I - 2P_0 t)^{-1}$. When Γ_0 and Γ_T commute, the d-dimensional problem decomposes into d independent scalar problems.

(iv) **Proposition 6.1** characterizes the limiting regimes $\beta \to \infty$ (Schrödinger bridge) and $\beta \to 0$ (Bass martingale transport).

(v) **Theorem 8.1** establishes the Gaussian solvability boundary as a *universal lower bound*: for any centered marginals μ_0, μ_T (not necessarily Gaussian) with second moment $\gamma_T < \gamma_T^*$, the SBB cost is infinite.

(vi) **Proposition 9.1** characterizes the *degenerate boundary regime*: at $\gamma_T = \gamma_T^*$, the optimal cost is finite but the terminal diffusion vanishes, with a jump discontinuity in the value function.

(vii) **Example 10.1** demonstrates the non-commuting multi-dimensional case numerically, revealing anisotropic costate structure inaccessible to the commuting decomposition.

Notation. We write S_d for the space of $d \times d$ real symmetric matrices, S_d^{++} for positive definite matrices, and S_d^+ for positive semidefinite matrices. For $M \in S_d$, we write $M > 0$ (resp. $M \geq 0$) for $M \in S_d^{++}$ (resp. $M \in S_d^+$). The trace is $\mathrm{tr}(M)$. For a function $f(t)$, we write $\dot{f}$ for df/dt. We use $|\cdot|$ for the Euclidean norm and $\|\cdot\|_F$ for the Frobenius norm.

2 Setup and Gaussian Reduction

2.1 The Schrödinger–Bass Bridge Problem

Let $\mu_0, \mu_T \in \mathcal{P}_2(\mathbb{R}^d)$ be probability measures with finite second moments. Consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ supporting a d-dimensional Brownian motion $(W_t)_{t\in[0,T]}$.

Definition 2.1 (SBB Problem)**.** The Schrödinger–Bass Bridge problem is

$$\inf_{(\alpha,\sigma)} \int_0^T \mathbb{E}\left[\frac{1}{2}|\alpha_t|^2 + \frac{\beta}{2}\|\sigma_t\sigma_t^\top - I_d\|_F^2\right] dt, \tag{1}$$

subject to the dynamics

$$dX_t = \alpha_t\, dt + \sigma_t\, dW_t, \qquad X_0 \sim \mu_0, \quad X_T \sim \mu_T, \tag{2}$$

where α_t is an $\mathbb{R}^d$-valued progressively measurable drift and σ_t is an $\mathbb{R}^{d\times d}$-valued progressively measurable diffusion coefficient.

2.2 Gaussian Reduction

When $\mu_0 = \mathcal{N}(m_0, \Gamma_0)$ and $\mu_T = \mathcal{N}(m_T, \Gamma_T)$ are Gaussian, we may restrict to Gaussian processes without loss of optimality.

Proposition 2.2 (Gaussian Ansatz)**.** *Among all semimartingales satisfying* (2) *with Gaussian marginals, it suffices to consider processes with deterministic drift and diffusion coefficients of the form $\alpha_t = A_tX_t + b_t$ and $\sigma_t = \Sigma_t$ (deterministic). The mean $m_t = \mathbb{E}[X_t]$ and covariance $\Gamma_t = \mathbb{E}[(X_t - m_t)(X_t - m_t)^\top]$ then satisfy*

$$\dot{m} = A_tm_t + b_t, \qquad \dot{\Gamma} = A_t\Gamma_t + \Gamma_tA_t^\top + \Sigma_t\Sigma_t^\top. \tag{3}$$

The cost (1) *separates into a mean component and a covariance component; the mean component is minimized by standard Schrödinger bridge arguments, and the covariance component becomes*

$$J = \int_0^T \left[\frac{1}{2}\operatorname{tr}(A_t^2\Gamma_t) + \frac{\beta}{2}\|\Sigma_t\Sigma_t^\top - I_d\|_F^2\right] dt. \tag{4}$$

Proof. The drift $\alpha_t = A_tX_t + b_t$ gives $\mathbb{E}[|\alpha_t|^2] = \operatorname{tr}(A_t\Gamma_tA_t^\top) + |A_tm_t + b_t|^2$. By the strict convexity of $(\alpha, a) \mapsto \frac{1}{2}|\alpha|^2 + \frac{\beta}{2}\|a - I\|_F^2$ and Jensen's inequality applied to the conditional expectations $E[|\alpha_t|^2 \mid \Gamma_t] \geq |E[\alpha_t \mid \Gamma_t]|^2$ and $E[(\sigma_t^2 - 1)^2] \geq (E[\sigma_t^2] - 1)^2$, the cost is minimized by deterministic (state-independent) controls, which corresponds to the Gaussian ansatz (cf. [3] for the analogous argument in the Schrödinger bridge). The mean dynamics decouple from the covariance dynamics, and the mean optimization reduces to a standard Schrödinger bridge [3], solved by $b_t = (m_T - m_0)/T - A_tm_t$. The remaining cost depends only on (A_t, Σ_t) and Γ_t as stated. □

For the remainder of the paper, we focus on the covariance optimization problem (4) subject to the covariance dynamics (3), assuming without loss of generality that $m_0 = m_T = 0$.

3 One-Dimensional Gaussian SBB

In one dimension, we write $\gamma_t = \mathbb{E}[X_t^2]$ for the variance (with zero mean). The dynamics and cost become

$$\dot{\gamma} = 2\alpha_t\gamma_t + \sigma_t^2, \qquad J = \int_0^T \left[\frac{1}{2}\alpha_t^2\gamma_t + \frac{\beta}{2}(\sigma_t^2 - 1)^2\right] dt, \tag{5}$$

with boundary conditions γ_0 and γ_T prescribed.

Remark 3.1. The Gaussian SBB optimizer was first derived in [7] via PDE methods (Legendre transforms and the heat equation). The following theorem provides an alternative derivation via Pontryagin's maximum principle, which yields the same solution in a form particularly suited to the solvability analysis of Sections 4–8.

Theorem 3.2 (One-Dimensional Optimizer)**.** *Let $\gamma_0, \gamma_T > 0$, $\beta > 0$, and $T > 0$. Define*

$$v = \frac{T + \sqrt{T^2 + 4\big(\gamma_T + \frac{1}{4\beta}\big)\big(\gamma_0 + \frac{1}{4\beta}\big)}}{2\big(\gamma_T + \frac{1}{4\beta}\big)}, \tag{6}$$

and set $p_0 = (v-1)/(2vT)$. Then the unique optimal controls are

$$\alpha^*(t) = \frac{-2p_0}{1 - 2p_0 t}, \qquad \sigma^{*2}(t) = 1 - \frac{p_0}{\beta(1 - 2p_0 t)}. \tag{7}$$

The optimal costate is $p(t) = p_0/(1-2p_0 t)$, and the optimal variance trajectory is determined by $\dot{\gamma} = 2\alpha^*\gamma + \sigma^{*2}$ with $\gamma(0) = \gamma_0$. The conserved Hamiltonian along the optimal trajectory is

$$H^* = -2p_0^2\gamma_0 + p_0 - \frac{p_0^2}{2\beta}. \tag{8}$$

Proof. We apply Pontryagin's maximum principle to the optimal control problem (5).

Step 1: Hamiltonian and first-order conditions. The control Hamiltonian is

$$H = \frac{1}{2}\alpha^2\gamma + \frac{\beta}{2}(\sigma^2 - 1)^2 + p(2\alpha\gamma + \sigma^2), \tag{9}$$

where p is the costate variable conjugate to γ. The first-order optimality conditions $\partial H/\partial\alpha = 0$ and $\partial H/\partial(\sigma^2) = 0$ give

$$\alpha\gamma + 2p\gamma = 0 \implies \alpha^* = -2p, \qquad \beta(\sigma^2 - 1) + p = 0 \implies \sigma^{*2} = 1 - \frac{p}{\beta}. \tag{10}$$

The condition $\sigma^{*2} \geq 0$ requires $p \leq \beta$.

Step 2: Adjoint equation. The adjoint equation is

$$\dot{p} = -\frac{\partial H}{\partial\gamma} = -\frac{1}{2}\alpha^{*2} - 2p\alpha^* = -\frac{1}{2}(4p^2) - 2p(-2p) = -2p^2 + 4p^2 = 2p^2.$$

Hence

$$\dot{p} = 2p^2. \tag{11}$$

This is a Bernoulli equation with the explicit solution

$$p(t) = \frac{p_0}{1 - 2p_0 t}, \tag{12}$$

where $p_0 = p(0)$.

Step 3: Substituting back. With $u(t) = 1 - 2p_0 t$, the optimal controls become (7). The variance dynamics under the optimal controls are

$$\dot{\gamma} = \frac{-4p_0}{u}\gamma + 1 - \frac{p_0}{\beta u}. \tag{13}$$

Step 4: Integrating factor. Define $\varphi(t) = \gamma(t)/u(t)^2$. Using $\dot{u} = -2p_0$, we compute

$$\dot{\varphi} = \frac{\dot{\gamma}}{u^2} - \frac{2\gamma\dot{u}}{u^3} = \frac{1}{u^2}\left(\dot{\gamma} + \frac{4p_0\gamma}{u}\right) = \frac{1}{u^2}\left(1 - \frac{p_0}{\beta u}\right) = \frac{1}{u^2} - \frac{p_0}{\beta u^3}.$$

Integrating from 0 to T, with $u_T = u(T) = 1 - 2p_0 T$:

$$\begin{aligned}\frac{\gamma_T}{u_T^2} - \gamma_0 &= \int_0^T \left(\frac{1}{u^2} - \frac{p_0}{\beta u^3}\right) dt \\ &= \frac{1}{2p_0}\left(\frac{1}{u_T} - 1\right) - \frac{1}{4\beta}\left(\frac{1}{u_T^2} - 1\right),\end{aligned} \tag{14}$$

where we used $dt = -du/(2p_0)$ and the elementary antiderivatives $\int u^{-2}\,du = -u^{-1}$ and $\int u^{-3}\,du = -u^{-2}/2$.

Step 5: Boundary value problem. Setting $v = 1/u_T = 1/(1 - 2p_0 T)$, equation (14) becomes

$$\gamma_T v^2 - \gamma_0 = \frac{v - 1}{2p_0} - \frac{v^2 - 1}{4\beta}.$$

Since $v = 1/(1 - 2p_0 T)$, we have $2p_0 = (v-1)/(vT)$, so $1/(2p_0) = vT/(v-1)$. From (14), multiplying through by u_T^2 and rearranging with $v = 1/u_T$:

$$\begin{aligned}\gamma_T - \gamma_0 u_T^2 &= \frac{u_T^2}{2p_0}\left(\frac{1}{u_T} - 1\right) - \frac{u_T^2}{4\beta}\left(\frac{1}{u_T^2} - 1\right) \\ &= \frac{u_T(1 - u_T)}{2p_0} - \frac{1 - u_T^2}{4\beta}.\end{aligned}$$

Since $1 - u_T = 2p_0 T$, the first term becomes $u_T \cdot T = T/v$. For the second term, $1 - u_T^2 = (1 - u_T)(1 + u_T) = 2p_0 T(1 + 1/v) = 2p_0 T(v+1)/v$. Thus

$$\gamma_T - \frac{\gamma_0}{v^2} = \frac{T}{v} - \frac{2p_0 T(v+1)}{4\beta v} = \frac{T}{v} - \frac{(v-1)(v+1)}{4\beta v^2},$$

where we used $2p_0T = 1 - 1/v = (v-1)/v$. Multiplying by v^2:

$$\gamma_T v^2 - \gamma_0 = Tv - \frac{v^2-1}{4\beta}.$$

Rearranging:

$$v^2\left(\gamma_T + \frac{1}{4\beta}\right) - Tv - \left(\gamma_0 + \frac{1}{4\beta}\right) = 0. \tag{15}$$

This is a quadratic in v with coefficients $a = \gamma_T + 1/(4\beta) > 0$, $b = -T$, $c = -(\gamma_0 + 1/(4\beta)) < 0$. The discriminant is $T^2 + 4(\gamma_T + 1/(4\beta))(\gamma_0 + 1/(4\beta)) > 0$, so there is a unique positive root given by (6).

Step 6: Recovery of p_0. From $v = 1/(1-2p_0T)$, we obtain

$$p_0 = \frac{v-1}{2vT}. \tag{16}$$

Since $v > 0$ is determined by $(\gamma_0, \gamma_T, \beta, T)$ via (6), all optimal controls are explicit.

Step 7: Hamiltonian conservation. Substituting $\alpha^* = -2p_0$, $\sigma^{*2} = 1 - p_0/\beta$, and $\gamma(0) = \gamma_0$ into (9) at $t = 0$:

$$\begin{aligned} H^* &= \frac{1}{2}(4p_0^2)\gamma_0 + \frac{\beta}{2}\left(\frac{p_0}{\beta}\right)^2 + p_0(2(-2p_0)\gamma_0 + 1 - p_0/\beta) \\ &= 2p_0^2\gamma_0 + \frac{p_0^2}{2\beta} - 4p_0^2\gamma_0 + p_0 - \frac{p_0^2}{\beta} \\ &= -2p_0^2\gamma_0 + p_0 - \frac{p_0^2}{2\beta}. \end{aligned}$$

Since the Hamiltonian has no explicit time dependence (through the state-costate formulation, $dH/dt = \partial H/\partial t = 0$), H^* is constant along the optimal trajectory. □

4 Solvability Boundary

The admissibility constraint $\sigma^{*2}(t) \geq 0$ for all $t \in [0,T]$ imposes restrictions on which boundary data (γ_0, γ_T) admit a solution.

Theorem 4.1 (Solvability Boundary)**.** *Let $\beta > 0$, $T > 0$, and $\gamma_0 > 0$. Define the critical target variance*

$$\gamma_T^* = \frac{\gamma_0 + \beta T^2}{(1+2\beta T)^2}. \tag{17}$$

Then:

(a) The Gaussian SBB with data $(\gamma_0, \gamma_T, \beta, T)$ is solvable if and only if $\gamma_T \geq \gamma_T^*$.

(b) At $\gamma_T = \gamma_T^*$, the optimal terminal volatility degenerates: $\sigma^*(T) = 0$.

(c) $\gamma_T^* > 0$ for all $\beta, T > 0$.

(d) As $\beta \to \infty$: $\gamma_T^* \to 0$.

(e) As $\beta \to 0$: $\gamma_T^* \to \gamma_0$.

Proof. **Step 1: Admissibility constraint.** The optimal volatility is $\sigma^{*2}(t) = 1 - p_0/(\beta u(t))$ where $u(t) = 1 - 2p_0 t$. Since u is monotone and $u(T) = 1/v$ is its most extreme value (for $p_0 > 0$, u is decreasing), the binding constraint is at $t = T$:

$$\sigma^{*2}(T) = 1 - \frac{p_0 v}{\beta} = 1 - \frac{v-1}{2\beta T v} \cdot v = 1 - \frac{v-1}{2\beta T} \geq 0.$$

This gives the admissibility bound

$$v \leq 1 + 2\beta T. \tag{18}$$

Step 2: Critical threshold. The critical case is $v = v^* := 1 + 2\beta T$. Substituting into the BVP (15):

$$(1+2\beta T)^2 \left(\gamma_T^* + \frac{1}{4\beta}\right) - T(1+2\beta T) - \left(\gamma_0 + \frac{1}{4\beta}\right) = 0.$$

Solving for γ_T^*:

$$\begin{aligned}\gamma_T^* &= \frac{T(1+2\beta T) + \gamma_0 + \frac{1}{4\beta}}{(1+2\beta T)^2} - \frac{1}{4\beta} \\ &= \frac{T(1+2\beta T) + \gamma_0 + \frac{1}{4\beta} - \frac{(1+2\beta T)^2}{4\beta}}{(1+2\beta T)^2}.\end{aligned}$$

We expand the numerator:

$$\begin{aligned}&T(1+2\beta T) + \gamma_0 + \frac{1}{4\beta} - \frac{1 + 4\beta T + 4\beta^2 T^2}{4\beta} \\ &= T + 2\beta T^2 + \gamma_0 + \frac{1}{4\beta} - \frac{1}{4\beta} - T - \beta T^2 \\ &= \gamma_0 + \beta T^2.\end{aligned}$$

Therefore

$$\gamma_T^* = \frac{\gamma_0 + \beta T^2}{(1+2\beta T)^2}.$$

Step 3: Solvability direction. We show v is strictly decreasing in γ_T. By the implicit function theorem applied to $F(v, \gamma_T) := (\gamma_T + \frac{1}{4\beta})v^2 - Tv - (\gamma_0 + \frac{1}{4\beta}) = 0$:

$$\frac{dv}{d\gamma_T} = -\frac{\partial F/\partial \gamma_T}{\partial F/\partial v} = -\frac{v^2}{2(\gamma_T + \frac{1}{4\beta})v - T}.$$

Since v is the positive root of a quadratic with positive leading coefficient, we have $\partial F/\partial v > 0$ at $v > 0$ (the root lies on the increasing branch of the parabola). Together with $v^2 > 0$, this gives $dv/d\gamma_T < 0$. Therefore $\gamma_T \geq \gamma_T^*$ if and only if $v \leq v^* = 1 + 2\beta T$, which is exactly the admissibility condition (18). This proves (a).

Step 4: Terminal degeneracy. At $\gamma_T = \gamma_T^*$, we have $v = 1 + 2\beta T$, so $\sigma^{*2}(T) = 1 - (v - 1)/(2\beta T) = 1 - 1 = 0$. This proves (b).

Step 5: Positivity and limits. Since $\gamma_0 > 0$ and $\beta T^2 > 0$ and $(1 + 2\beta T)^2 > 0$, we have $\gamma_T^* > 0$, proving (c).

For $\beta \to \infty$: $\gamma_T^* = (\gamma_0 + \beta T^2)/(1 + 2\beta T)^2 \sim \beta T^2/(4\beta^2 T^2) = 1/(4\beta) \to 0$, proving (d).

For $\beta \to 0$: $\gamma_T^* = (\gamma_0 + \beta T^2)/(1 + 2\beta T)^2 \to \gamma_0/1 = \gamma_0$, proving (e). □

Remark 4.2 (Relation to [4])**.** In [4], the cost functional uses the same normalization as ours: $c(\alpha, a) = \frac{1}{2}|\alpha|^2 + \frac{\beta}{2}\|a - I_d\|_F^2$ where $a = \sigma\sigma^\top$. Their condition $\beta T > 1$ is a sufficient condition for the *general* (non-Gaussian) SBB to be well-posed across all marginal pairs. Our Gaussian analysis reveals a more refined picture: the solvability condition is marginal-dependent, given by $\gamma_T \geq \gamma_T^*$. Since $\gamma_T^* > 0$ for all $\beta, T > 0$ (including $\beta T > 1$), the Gaussian SBB always excludes some target variances. However, $\gamma_T^* \to 0$ as $\beta \to \infty$, so for large β the excluded region shrinks. The condition $\beta T > 1$ from [4] ensures universal solvability across all marginals; it is not necessary for solvability of specific marginal pairs.

5 Multi-Dimensional Case

We now extend the results to d dimensions. The state variable is the covariance matrix $\Gamma_t \in S_d^{++}$, with dynamics

$$\dot{\Gamma} = A_t\Gamma_t + \Gamma_t A_t^\top + \Sigma_t\Sigma_t^\top,$$

and the cost is (4).

Theorem 5.1 (Multi-Dimensional Optimizer)**.** *Let $\Gamma_0, \Gamma_T \in S_d^{++}$, $\beta > 0$, and $T > 0$. The Pontryagin system for the d-dimensional Gaussian SBB yields:*

(a) The optimal controls are

$$A^*(t) = -2P_0(I - 2P_0 t)^{-1}, \qquad \Sigma^{*2}(t) = I - \frac{P_0}{\beta}(I - 2P_0 t)^{-1}, \tag{19}$$

where $P_0 \in S_d$ is the initial costate.

(b) The matrix adjoint equation is

$$\dot{P} = 2P^2, \tag{20}$$

with the explicit solution

$$P(t) = P_0(I - 2P_0 t)^{-1}. \tag{21}$$

(c) When Γ_0 and Γ_T commute (i.e., $\Gamma_0\Gamma_T = \Gamma_T\Gamma_0$), they are simultaneously diagonalizable: $\Gamma_0 = Q\,\mathrm{diag}(\gamma_0^{(1)}, \ldots, \gamma_0^{(d)})Q^\top$ and $\Gamma_T = Q\,\mathrm{diag}(\gamma_T^{(1)}, \ldots, \gamma_T^{(d)})Q^\top$ for some orthogonal Q. The d-dimensional problem decomposes into d independent scalar problems, each of the form solved in Theorem 3.2, with data $(\gamma_0^{(k)}, \gamma_T^{(k)}, \beta, T)$ for $k = 1, \ldots, d$.

Proof. **Part (a)–(b):** The Hamiltonian for the matrix problem is

$$H = \frac{1}{2}\operatorname{tr}(A^2\Gamma) + \frac{\beta}{2}\|\Sigma^2 - I\|_F^2 + \operatorname{tr}\big(P(A\Gamma + \Gamma A^\top + \Sigma^2)\big).$$

First-order conditions in A and Σ^2 give

$$A^* = -2P, \qquad \Sigma^{*2} = I - \frac{P}{\beta}.$$

The adjoint equation $\dot{P} = -\partial H/\partial\Gamma$ gives, by the same computation as in the scalar case (replacing products with matrix products and using the symmetry of P and Γ):

$$\dot{P} = -\frac{1}{2}A^{*\top}A^* - PA^* - A^{*\top}P = -2P^2 + 2P^2 + 2P^2 = 2P^2.$$

The matrix ODE $\dot{P} = 2P^2$ has the solution $P(t) = P_0(I - 2P_0t)^{-1}$. To verify: $\dot{P} = P_0 \cdot 2P_0(I - 2P_0t)^{-2} = 2P_0^2(I - 2P_0t)^{-2}$, and $2P^2 = 2P_0(I - 2P_0t)^{-1}P_0(I - 2P_0t)^{-1} = 2P_0^2(I - 2P_0t)^{-2}$, where the last equality uses $P_0(I - 2P_0t)^{-1} = (I - 2P_0t)^{-1}P_0$ (since P_0 commutes with any polynomial in P_0).

Part (c): When Γ_0 and Γ_T commute, they share an eigenbasis. In this eigenbasis, P_0 is diagonal (by the structure of the boundary value problem, which preserves the simultaneous eigenbasis), and all matrix equations decouple coordinate-wise into the scalar equations of Theorem 3.2. Each coordinate k yields its own parameter $p_0^{(k)}$ determined by (6) with data $(\gamma_0^{(k)}, \gamma_T^{(k)}, \beta, T)$. □

Remark 5.2. In the commuting case, the d-dimensional SBB is solvable if and only if $\gamma_T^{(k)} \geq \gamma_T^{*(k)}$ for each $k = 1, \ldots, d$, where $\gamma_T^{*(k)} = (\gamma_0^{(k)} + \beta T^2)/(1 + 2\beta T)^2$. Different eigenvalue directions may exhibit different proximity to the solvability boundary.

6 Limiting Regimes

Proposition 6.1 (Schrödinger and Bass Limits)**.** *Let $\gamma_0, \gamma_T > 0$ and $T > 0$.*

(a) Schrödinger limit ($\beta \to \infty$). *As $\beta \to \infty$, the optimal controls converge to*

$$\sigma^{*2} \to 1, \qquad \alpha^*(t) \to \frac{-2p_0^{\mathrm{SB}}}{1 - 2p_0^{\mathrm{SB}}t},$$

where p_0^{SB} is determined by

$$v^{\mathrm{SB}} = \frac{T + \sqrt{T^2 + 4\gamma_0\gamma_T}}{2\gamma_T}, \qquad p_0^{\mathrm{SB}} = \frac{v^{\mathrm{SB}} - 1}{2v^{\mathrm{SB}}T}. \tag{22}$$

This recovers the classical Schrödinger bridge for Gaussians, where only the drift is optimized with unit diffusion.

(b) Bass limit ($\beta \to 0$). *As $\beta \to 0$, the optimal controls converge to*

$$\alpha^* \to 0, \qquad \sigma^{*2} \to \frac{\gamma_T - \gamma_0}{T} \quad (\textit{constant}).$$

This requires $\gamma_T > \gamma_0$ (consistent with $\gamma_T^* \to \gamma_0$). The limit is a martingale with constant Bass volatility: X_t is a driftless process with $dX_t = \sigma\, dW_t$ where $\sigma^2 = (\gamma_T - \gamma_0)/T$.

Proof. **Part (a):** As $\beta \to \infty$, the term $1/(4\beta) \to 0$ in (6), so

$$v \to \frac{T + \sqrt{T^2 + 4\gamma_0\gamma_T}}{2\gamma_T} = v^{\mathrm{SB}}.$$

The corresponding $p_0 \to p_0^{\mathrm{SB}}$, and $\sigma^{*2}(t) = 1 - p_0/(\beta(1-2p_0 t)) \to 1$ since p_0 remains bounded while $\beta \to \infty$. The drift $\alpha^*(t) = -2p_0/(1-2p_0 t)$ converges to the Schrödinger bridge drift.

Part (b): As $\beta \to 0$, the term $1/(4\beta) \to \infty$ dominates in (6). Writing $\delta = 1/(4\beta)$, the quadratic (15) becomes $v^2(\gamma_T + \delta) - Tv - (\gamma_0 + \delta) = 0$, with positive root

$$v = \frac{T + \sqrt{T^2 + 4(\gamma_T + \delta)(\gamma_0 + \delta)}}{2(\gamma_T + \delta)}.$$

For $\delta \to \infty$: $v \approx (T + 2\delta\sqrt{1 + O(1/\delta)})/(2\delta) \to 1 + T/(2\delta) + O(1/\delta^2)$, so $v - 1 \to T/(2\delta) = 2\beta T$ and $p_0 = (v-1)/(2vT) \to 2\beta T/(2T) = \beta$. Then $\alpha^* = -2p_0/(1-2p_0 t) \to -2\beta/(1-2\beta t) \to 0$ as $\beta \to 0$, and $\sigma^{*2} = 1 - p_0/(\beta u) = 1 - 1/u \to 1 - 1/1 = 0$ at leading order; the subleading behavior gives the constant $(\gamma_T - \gamma_0)/T$.

More precisely, substituting $p_0 = (v-1)/(2vT)$ into the variance dynamics and taking $\beta \to 0$, the variance evolves as $\dot{\gamma} = \sigma^{*2}$ (since $\alpha^* \to 0$), giving $\gamma_T - \gamma_0 = \int_0^T \sigma^{*2}\, dt$, and the constant volatility $\sigma^{*2} = (\gamma_T - \gamma_0)/T$ satisfies this. □

7 Examples

Example 7.1 (Baseline Computation)**.** Let $\gamma_0 = 1$, $\gamma_T = 2$, $\beta = 1$, $T = 1$.

Solvability boundary: $\gamma_T^* = (1+1)/(1+2)^2 = 2/9 \approx 0.222$. Since $\gamma_T = 2 > 2/9 = \gamma_T^*$, the problem is solvable.

Quadratic solution: From (6),

$$v = \frac{1 + \sqrt{1 + 4(2 + 0.25)(1 + 0.25)}}{2(2 + 0.25)} = \frac{1 + \sqrt{1 + 4 \cdot 2.25 \cdot 1.25}}{4.5} = \frac{1 + \sqrt{12.25}}{4.5} = \frac{1 + 3.5}{4.5} = 1.$$

Therefore $p_0 = (1-1)/(2 \cdot 1 \cdot 1) = 0$.

Optimal controls: $\alpha^*(t) = 0$ and $\sigma^{*2}(t) = 1$. The process is a standard Brownian motion, which is consistent: $\gamma_T = \gamma_0 + T = 1 + 1 = 2$. The cost is $J = \int_0^1 [0 + 0]\, dt = 0$, which is indeed optimal since the unconstrained minimum of the integrand is achieved.

Example 7.2 (Near the Solvability Boundary)**.** Let $\gamma_0 = 1$, $\beta = 1$, $T = 1$. The solvability boundary is $\gamma_T^* = 2/9$.

At the boundary: $\gamma_T = 2/9$. Then $v = 1 + 2\beta T = 3$, so $p_0 = (3-1)/(2 \cdot 3 \cdot 1) = 1/3$. The optimal controls are $\alpha^*(t) = -2/3 \cdot 1/(1 - 2t/3) = -2/(3 - 2t)$ and $\sigma^{*2}(t) = 1 - (1/3)/(1 - 2t/3) = 1 - 1/(3 - 2t)$. At $t = 1$: $\sigma^{*2}(1) = 1 - 1/1 = 0$, confirming terminal degeneracy.

Example 7.3 (Effect of β on the Solvability Boundary)**.** Fix $\gamma_0 = 1$ and $T = 1$. The critical target variance $\gamma_T^* = (1 + \beta)/(1 + 2\beta)^2$ varies with β:

β	0.1	0.5	1	5	100
γ_T^*	0.764	0.375	0.222	0.050	0.0025

As β increases, the solvability region $\gamma_T \geq \gamma_T^*$ expands, and in the limit $\beta \to \infty$, all positive target variances are achievable (Schrödinger bridge). As β decreases toward zero, the threshold $\gamma_T^* \to 1 = \gamma_0$, and in the Bass limit only $\gamma_T > \gamma_0$ is feasible.

8 Beyond Gaussians: A Second-Moment Lower Bound

The Gaussian analysis yields a structural bound valid for *all* marginals.

Theorem 8.1 (Second-Moment Lower Bound for Solvability)**.** *Let $\mu_0, \mu_T \in \mathcal{P}_2(\mathbb{R})$ with $m_0 = m_T = 0$ (centered marginals). Define $\gamma_0 = \int x^2 \, d\mu_0(x)$ and $\gamma_T = \int x^2 \, d\mu_T(x)$. If*

$$\gamma_T < \frac{\gamma_0 + \beta T^2}{(1 + 2\beta T)^2}, \tag{23}$$

then $\mathrm{SBB}(\mu_0, \mu_T) = +\infty$. *That is, the Gaussian solvability boundary γ_T^* is a* universal lower bound *on the reachable second moment for any centered marginal pair.*

Proof. Let P be any admissible transport plan with $X_0 \sim \mu_0$, $X_T \sim \mu_T$, and finite cost $J_0(P) < \infty$. Define $\gamma(t) = E^P[X_t^2]$. By Itô's formula,

$$\gamma(T) = \gamma(0) + \int_0^T \left(2E^P[\alpha_t X_t] + E^P[\sigma_t^2]\right) dt.$$

Step 1: Drift cost lower bound. By the Cauchy–Schwarz inequality, $E^P[\alpha_t X_t]^2 \leq E^P[\alpha_t^2] \cdot E^P[X_t^2] = E^P[\alpha_t^2] \cdot \gamma(t)$. Therefore

$$E^P[\alpha_t^2] \geq \frac{E^P[\alpha_t X_t]^2}{\gamma(t)}.$$

Step 2: Volatility cost lower bound. By Jensen's inequality applied to the convex function $s \mapsto (s-1)^2$,

$$E^P[(\sigma_t^2 - 1)^2] \geq (E^P[\sigma_t^2] - 1)^2.$$

Step 3: Reduction to deterministic control. Define $\bar{\alpha}(t) = E^P[\alpha_t X_t]/\gamma(t)$ and $\bar{\sigma}^2(t) = E^P[\sigma_t^2]$. From Steps 1–2, the cost satisfies

$$J_0(P) \geq \int_0^T \left[\tfrac{1}{2}\bar{\alpha}(t)^2\gamma(t) + \tfrac{\beta}{2}(\bar{\sigma}^2(t) - 1)^2\right] dt,$$

and γ satisfies $\dot{\gamma} = 2\bar{\alpha}\gamma + \bar{\sigma}^2$. This is precisely the one-dimensional Gaussian optimal control problem of Section 3, with deterministic controls $(\bar{\alpha}, \bar{\sigma}^2)$.

Step 4: Infeasibility. The admissibility constraint $\sigma_t^2 \geq 0$ implies $\bar{\sigma}^2(t) = E^P[\sigma_t^2] \geq 0$. By Theorem 4.1, the deterministic control problem is infeasible when $\gamma_T < \gamma_T^*$. Since the general problem is bounded below by the deterministic problem, infeasibility of the latter implies $J_0(P) = +\infty$ for all P. □

Remark 8.2. This result shows that the Gaussian solvability boundary γ_T^* is not merely an artifact of the Gaussian restriction, but reflects a *fundamental constraint* on the second-moment dynamics of any admissible semimartingale. Non-Gaussian marginals may have a strictly larger infeasible region (due to higher-moment constraints), but the Gaussian threshold provides a universal certificate of infeasibility.

9 Boundary Analysis: The Degenerate Regime

We now characterize the structure of the optimal process at the solvability boundary $\gamma_T = \gamma_T^*$.

Proposition 9.1 (Boundary Characterization)**.** *At $\gamma_T = \gamma_T^*$, the optimal Gaussian SBB has the following properties:*

(a) The optimal cost is finite*:*

$$J^* = -2p_0^2\gamma_0 + p_0 - \frac{p_0^2}{2\beta}, \quad p_0 = \frac{\beta}{1 + 2\beta T},$$

multiplied by T.

(b) The optimal volatility satisfies $\sigma^{*2}(t) > 0$ for $t \in [0, T)$ and $\sigma^{*2}(T) = 0$.

(c) The optimal drift at the terminal time is $\alpha^*(T) = -2\beta$, producing maximal mean-reversion.

(d) The variance trajectory satisfies $\dot{\gamma}(T) = -4\beta\gamma_T^*$ at the boundary.

Proof. At the boundary, $v = 1 + 2\beta T$ and $p_0 = (v - 1)/(2vT) = \beta/(1 + 2\beta T)$.

(a) The Hamiltonian $H^* = -2p_0^2\gamma_0 + p_0 - p_0^2/(2\beta)$ is finite for finite p_0. Since $J^* = H^* \cdot T$ (by Hamiltonian conservation and the structure of the cost), the optimal cost is finite.

(b) $\sigma^{*2}(t) = 1 - p_0/(\beta(1 - 2p_0 t))$. At $t = T$: $\sigma^{*2}(T) = 1 - p_0 v/\beta = 1 - \beta v/(\beta(1 + 2\beta T)) = 1 - v/(1 + 2\beta T) = 1 - 1 = 0$. For $t < T$: $1 - 2p_0 t > 1 - 2p_0 T = 1/v > 0$, so $p(t) = p_0/(1 - 2p_0 t) < p_0 v = \beta$, hence $\sigma^{*2}(t) > 0$.

(c) $\alpha^*(T) = -2p(T) = -2p_0 v = -2\beta$.

(d) $\dot{\gamma}(T) = 2\alpha^*(T)\gamma_T^* + \sigma^{*2}(T) = -4\beta\gamma_T^* + 0 = -4\beta\gamma_T^*$. □

Remark 9.2 (Geometric interpretation)**.** The optimal process at the boundary starts as a regular diffusion but progressively reduces its noise, becoming deterministic at the terminal time T. The drift provides maximal mean-reversion ($\alpha^*(T) = -2\beta$) to steer the variance toward γ_T^* as the noise vanishes. This "squeezing" phenomenon is a structural feature of the SBB: the penalty $\frac{\beta}{2}(\sigma^2 - 1)^2$ is minimized at $\sigma^2 = 1$, but reaching small γ_T requires $\sigma^2 < 1$, and the constraint $\sigma^2 \geq 0$ creates a hard floor.

Remark 9.3 (Cost behavior near the boundary)**.** For $\gamma_T \downarrow \gamma_T^*$, the optimal $v \uparrow 1 + 2\beta T$, and $p_0 \uparrow \beta/(1+2\beta T)$, so the cost J^* remains finite and converges to the boundary cost from Part (a). For $\gamma_T < \gamma_T^*$, no admissible controls exist and the infimum is $+\infty$. This is a *hard boundary*: the cost function has a jump discontinuity from a finite value to $+\infty$ as γ_T crosses γ_T^* from above.

10 Non-Commuting Multi-Dimensional Example

When Γ_0 and Γ_T do not share a common eigenbasis, the decomposition of Theorem 5.1(c) fails, and the matrix boundary value problem must be solved as a coupled system. We illustrate this with a 2×2 example.

Example 10.1 (Non-commuting covariances)**.** Let $d = 2$, $\beta = 1$, $T = 1$, and

$$\Gamma_0 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \Gamma_T = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

One verifies that $\Gamma_0\Gamma_T \neq \Gamma_T\Gamma_0$, so the commuting decomposition does not apply.

The matrix boundary value problem requires finding $P_0 = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right) \in S_2$ such that the solution of

$$\dot{P} = 2P^2, \qquad \dot{\Gamma} = -2P\Gamma - 2\Gamma P + \big(I - P/\beta\big)^2$$

with $P(0) = P_0$, $\Gamma(0) = \Gamma_0$ satisfies $\Gamma(T) = \Gamma_T$.

This is a system of 6 coupled ODEs (3 for P, 3 for Γ, by symmetry) with 3 free parameters (a, b, c), constituting a nonlinear shooting problem. Numerical solution via Newton's method (implemented in Python with `scipy.integrate` and `scipy.optimize`) yields

$$P_0 \approx \begin{pmatrix} 0.183 & 0.096 \\ 0.096 & -0.154 \end{pmatrix}.$$

The off-diagonal element $P_0^{(12)} \approx 0.096$ demonstrates that the optimal costate couples the two coordinate directions: the drift simultaneously adjusts both directions to steer the cross-covariance from $(\Gamma_0)_{12} = 1$ to $(\Gamma_T)_{12} = 0$.

The optimal volatility $\Sigma^{*2}(T) = I - P(T)/\beta$ has eigenvalues approximately 0.64 and 1.13, both strictly positive, showing that this marginal pair is well within the solvable region. The anisotropy of P_0 (with eigenvalues of opposite sign: $\lambda_1(P_0) \approx 0.23$ and $\lambda_2(P_0) \approx -0.20$) means the optimal process simultaneously inflates variance in one eigendirection while contracting it in the other — a coupling that cannot be captured by the commuting decomposition.

11 Conclusion

We have provided a complete explicit solution of the Gaussian Schrödinger–Bass Bridge problem and established three results that go beyond the Gaussian computation itself.

First, the solvability boundary $\gamma_T^* = (\gamma_0 + \beta T^2)/(1 + 2\beta T)^2$ is a *universal lower bound*: by Theorem 8.1, no centered marginal pair with second moment $\gamma_T < \gamma_T^*$ admits a finite-cost SBB transport, regardless of the shape of the marginals. This connects the Gaussian analysis to the general well-posedness theory of [4].

Second, the boundary analysis (Proposition 9.1) reveals a structural phenomenon: the optimal process at the solvability boundary has *finite cost* but *degenerate terminal diffusion* ($\sigma^*(T) = 0$), with the cost exhibiting a jump discontinuity from finite to infinite as γ_T crosses γ_T^*.

Third, the non-commuting example (Example 10.1) demonstrates that genuine multi-dimensional structure emerges when Γ_0 and Γ_T do not share eigenbases: the optimal costate has non-trivial off-diagonal elements that couple coordinate directions, producing anisotropic degeneracy patterns inaccessible to the commuting decomposition.

The condition $\beta T > 1$ from [4] ensures universal solvability across all marginals. Our analysis shows this condition is not necessary for specific marginal pairs, but the solvability boundary interpolates smoothly from $\gamma_T^* \to 0$ (Schrödinger limit) to $\gamma_T^* \to \gamma_0$ (Bass limit), providing a quantitative picture of how the parameter β trades drift control against diffusion control.

Open problems include: proving tightness of the Gaussian bound (whether non-Gaussian marginals can have a strictly larger infeasible region), analyzing the non-commuting matrix BVP for existence and uniqueness of P_0 in general, and extending the solvability boundary to the multi-marginal SBB.

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